

Lesson 4.2: Bayes Decisions

Decisions with likelihoods

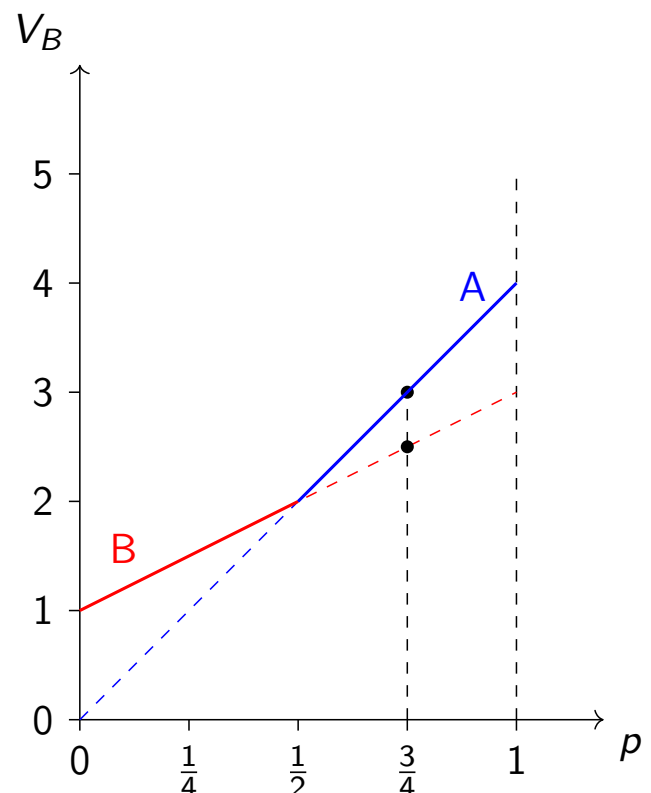
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Bayes decision rule

	$\frac{3}{4}$ f	$\frac{1}{4}$ $\bar{f}$	V_B
A	4	0	3.0
B	3	1	2.5

Probability dependence,
where $p = P(f)$:

	p f	$1 - p$ $\bar{f}$	V_B
A	4	0	$4p$
B	3	1	$1 + 2p$



Bayes indifference curves

General action, X:

	p f	$1 - p$ $\bar{f}$	V_B
A	4	0	$4p$
B	3	1	$1 + 2p$
X	v_1	v_2	$pv_1 + (1 - p)v_2$

